

Introduction to Machine Learning Methods

Computational learning agents

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Course summary

- Instructor: Haris Gačanin
- Assistant: Mohsen Pourghasemian, Firooz Bashashi Saghezchi
- Structure of the course
 - 2h lectures
 - 1h exercise (laboratory)
- Final exam: written (coding assignments are 10% of total grade)
- Lecture slides, research papers, additional material (provided electronically as needed)
- Textbook examples:
 - S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, Prentice Hall, Third Edition.

Course outline

- Introduction
- Supervised Learning
- Unsupervised learning
- Deep neural networks
- Computational agents
- Reinforcement learning
- Deep reinforcement learning

Goals of this course

- Understand differences between AI and ML
- What kind of problems we can solve with AI/ML
- Understand the requirements and limitations of AI/ML supported systems
- Compare different AI/ML designs and their applications
- Describe the most representative implementation algorithms



Introduction

Before we start

- What is a complex number?
- What is probability distribution?
- What is vector addition?
- What is quadratic form?
- What is gradient?

Outline of this lecture

- Type of agents
- Agent environments

Reference: Chapter 2

- **AI** enables systems to carry out full cognitive tasks in a way that humans consider intelligent
 - All of the characteristics of human intelligence: planning, perceiving, reasoning, learning, and problem solving
 - Design of intelligent agent

- **AI** enables systems to carry out full cognitive tasks in a way that humans consider intelligent
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- **ML** enables systems to learn with little or no guidance at all.
 - ML performs a range of human cognitive abilities -
Learning discipline of AI

What is intelligence?

What is learning?

Intelligence → AI

Learning → ML

AI → Autonomy

ML → Automation

- What is Automation System?

Understand now,

Enhance current decision

- What is **Autonomous System**?

Understand **now**,

Learn from past,

Enhance **future** decisions

Traditional optimization vs. generalization vs. acting

Given input x , solve $\omega^* = \arg \min_{\omega} f_x(\omega)$

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Iterative algorithm

```
while < Condition >  
     $\omega^{(n+1)} = T(\omega^n)$   
end  
output:  $\omega^*$ 
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Learning by examples

$$h_x(\omega) \approx \hat{f}_x$$

Learnt a priory (offline)

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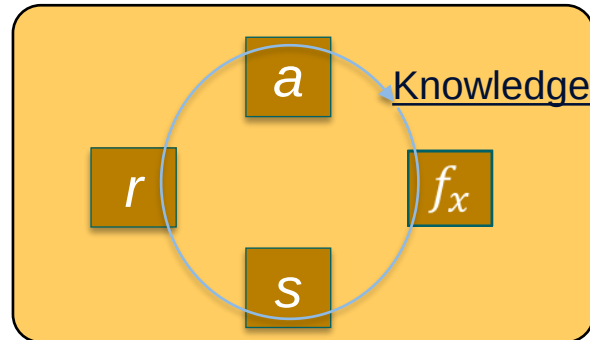
Learning by examples

$$h_x(\omega) \approx \hat{f}_x$$

Data

Learnt a priory (offline)

Learning to act



Applications

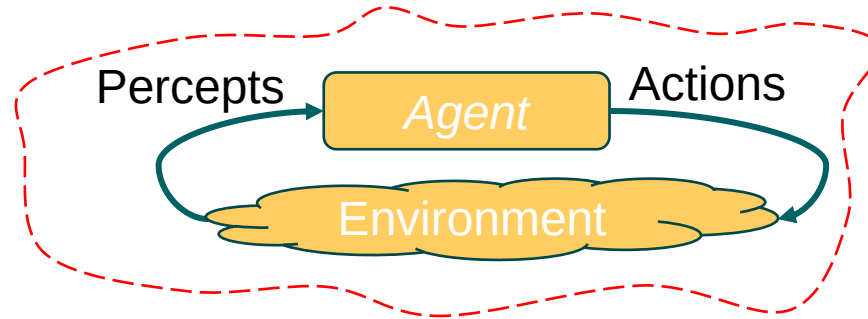
- **Manufacturing**
 - Tesla's factory comprises of more than 160 robots that do work on cars to reduce the risk of any defects
- **FinTech**
 - Evaluating strategies in stock market trading where Q-Learning algorithm is able to learn an optimal trading strategy with one simple instruction; maximize the value of our portfolio.
- **Inventory/Delivery Management**
 - Reducing transit time (smooth material flow) and minimize costs for stocking, retrieving products in the warehouse for optimizing space utilization and warehouse operations.
- **Social networking**
 - Create a social structure made up of a set of individuals or organizations or a combination of both
- **Power systems**
 - Assess the security of the electric power systems and to enhance Microgrid performance



Agents and Environments

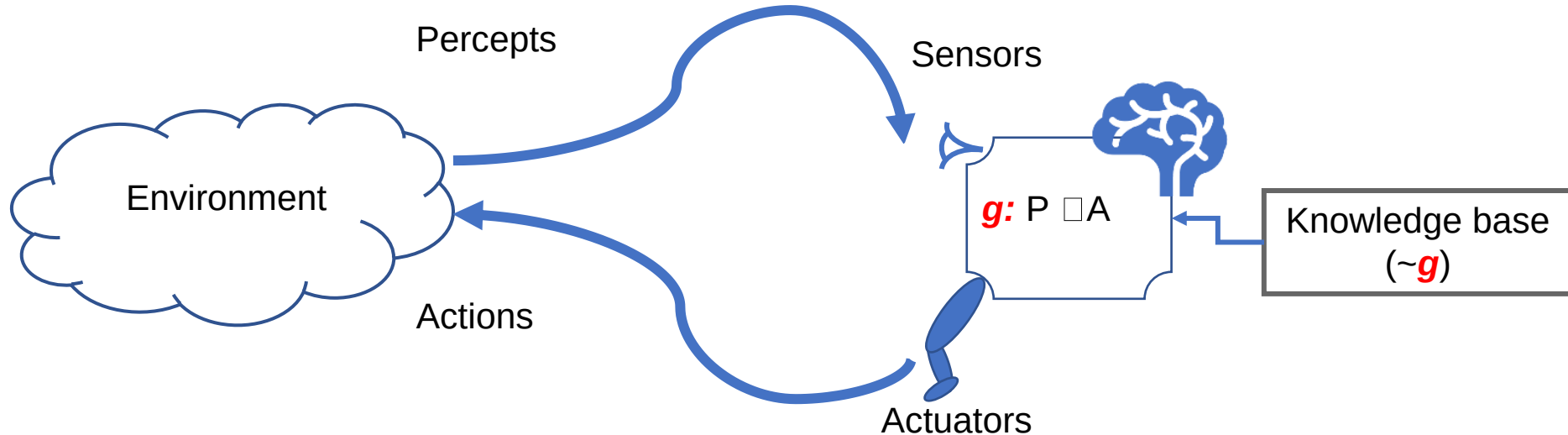
Computational (intelligent) agent

- An agent is anything that perceives its environment through sensors and acts upon that environment through actuators
 - **Human agent:** eyes, ears, and other organs for sensors; hands, legs, mouth, and other body parts for actuators
 - **Robotic agent:** cameras and infrared range finders for sensors; various motors for actuators



Agent and Environment

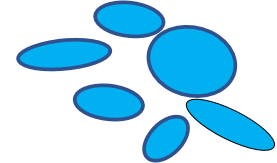
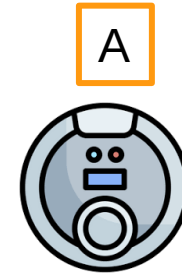
- Agent is autonomous to the extent that its behavior is determined by its own experience
- Initially the agent acts randomly unless the designer built-in some knowledge.



Example: Vacuum-cleaner world

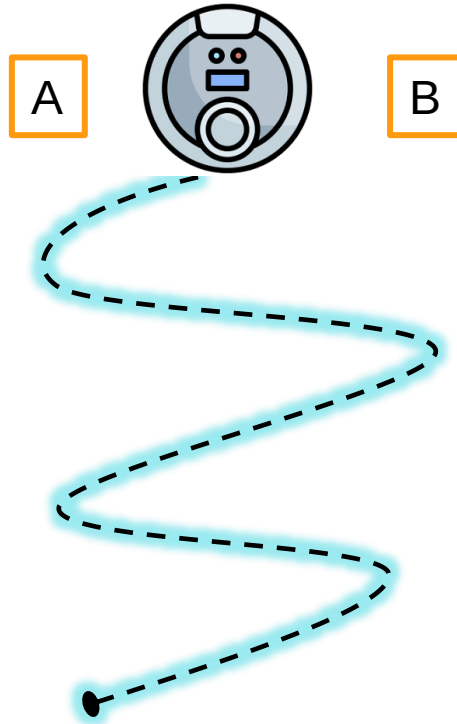
Vacuum-cleaner / state space	
Sensing percepts	[Location, Content] e.g., [<i>A</i> , <i>Not Dirty</i>]

Vacuum-cleaner / action space	
Actions	[Move, Clean] e.g., [<i>Right</i> , <i>On</i>]



Example: A vacuum-cleaner agent

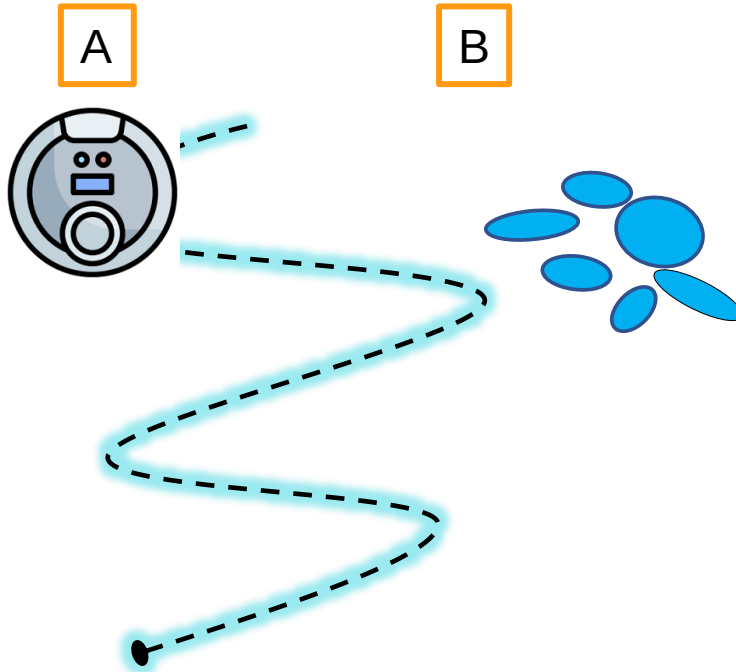
- What is the **correct** function g ? Can it be implemented in a small agent program?



Sensing	Action
?	?

Example: A vacuum-cleaner agent

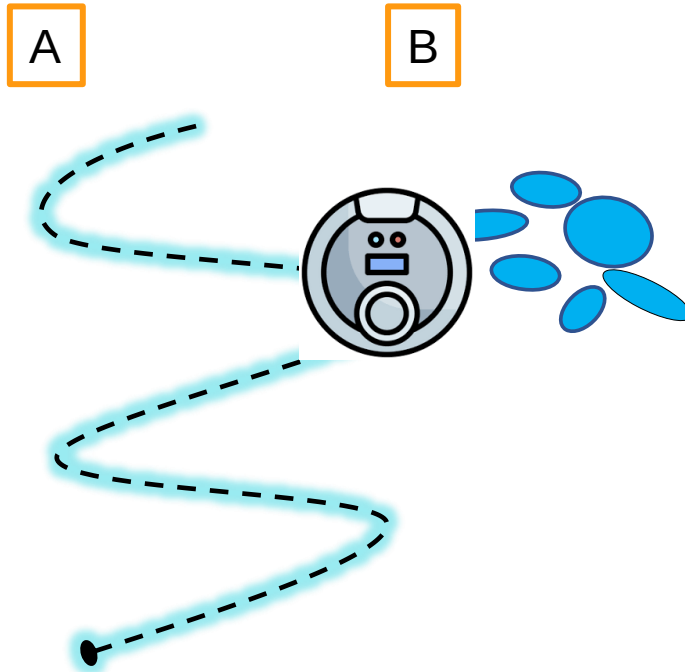
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Sensing	Action
[A, Clean]	Right

Example: A vacuum-cleaner agent

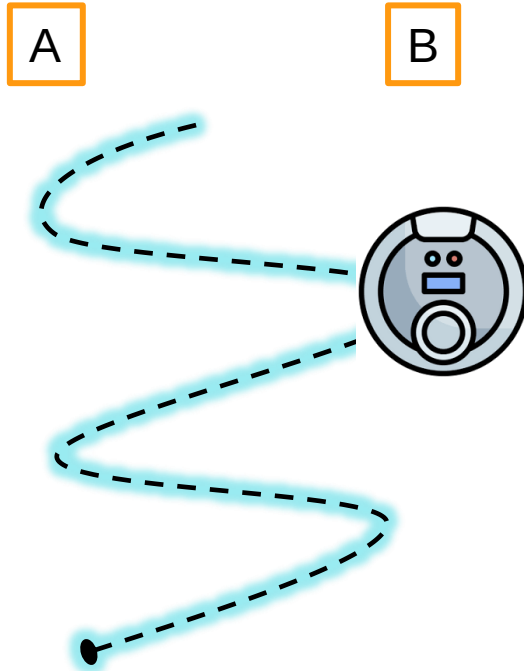
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Sensing	Action
[A, Clean]	Right
[B, Dirty]	On

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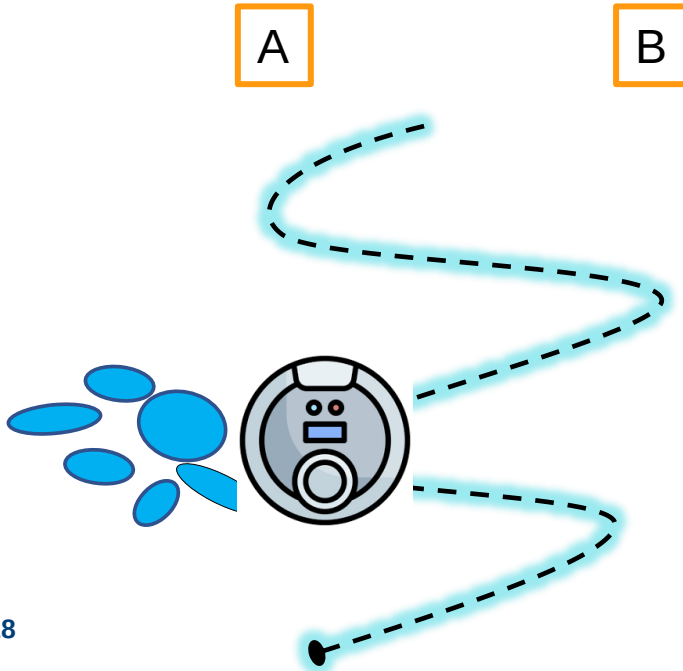
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Sensing	Action
[A, Clean]	Right
[B, Dirty]	On
[B, Clean]	Left

Example: A vacuum-cleaner agent

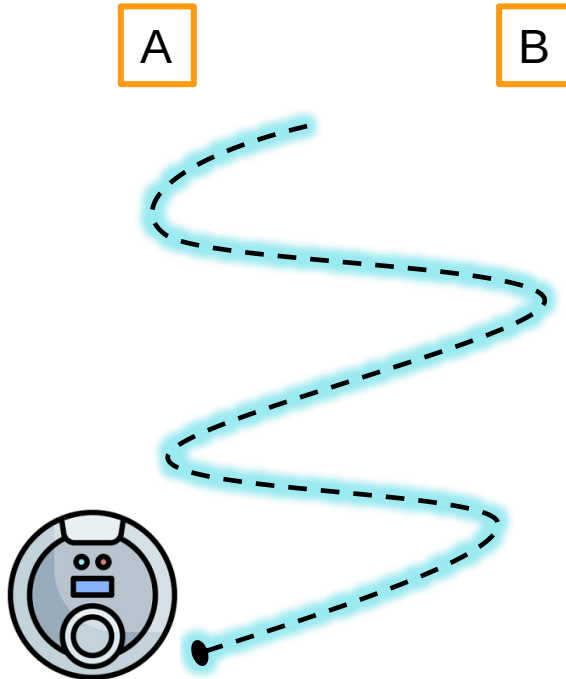
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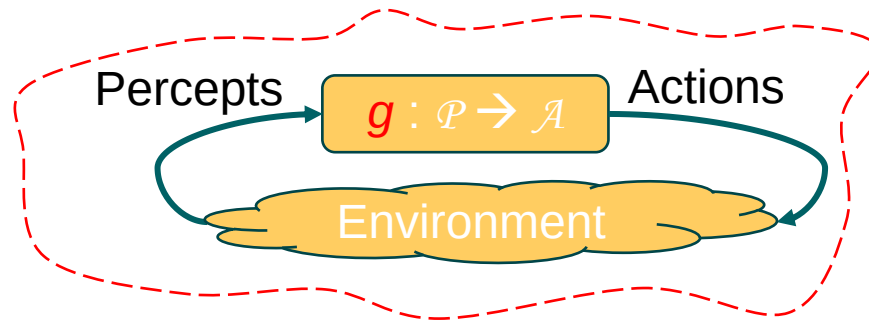
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Sensing	g	Action
[A, Clean]		Right
[A, Dirty]		On
[B, Clean]		Left
[A, Dirty]		On
⋮		⋮
[A, Clean]		Right
[B, Clean]		STOP

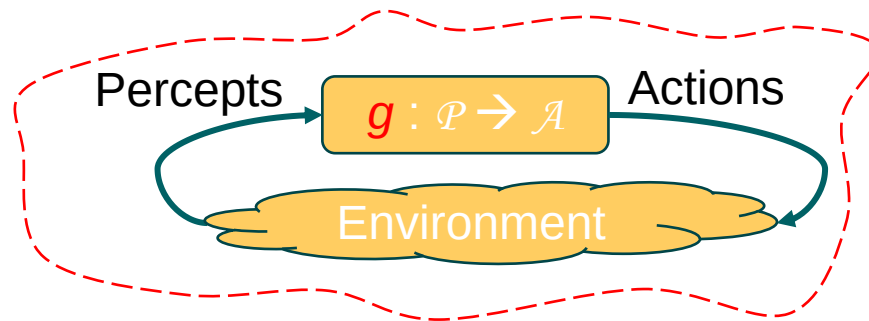
Definitions: Rational agents (1/2)

- **Rational agent** should strive to “do the right thing” based on what it perceives and the actions it can perform whatever built-in knowledge the agent has
 - Finds an action that is expected to maximize its performance measure.
- **Rationality is distinct** from omniscience (all-knowing with infinite knowledge) because
 - Percepts may/do not supply all relevant information.
- **The right action:** is the one that will cause the agent to be **most successful**



Definitions: Rational agents (2/2)

- Agents can perform actions **in order to modify future** percepts so as to obtain useful information.
 - Information gathering, exploration.
- Rational agent is **not always successful**.
- Performance measure**: An objective criterion for success of an agent's behavior
 - e.g., performance measure of a vacuum-cleaner agent may be the amount of cleaned dirt, amount of time taken, amount of electricity consumed, amount of noise generated, etc.



Task Environments

Partially observable problem (1/2)



Can we select
a route from
Melbourne to
Aachen?

Partially observable problem



How about
now?

Fully observable vs. partially observable

- Sensors give the agent an access to the **complete state of the environment** at each point in time ($O_t = S_t^{agent} = S_t^{envir.}$) – Markov decision process
 - State is relevant to the choice of action depending on a performance measure
- An environment may be **partially observable** due to noisy and inaccurate sensors
- Parts of the states are missing from sensor data leading to **partially observable** environment
- Without sensors, an environment is **unobservable**
 - Like the **test data set**, which is unseen

Deterministic vs. stochastic

- In Deterministic, the **next state** of the environment is **completely determined** by the **current state and the taken action**.
- In stochastic, the next state cannot be completely determined by the current state and taken action.

Episodic vs. sequential

- Episodic, the agent's experience is divided into atomic episodes
 - (consisting of the agent perceiving and then performing a single action)
 - The choice of action in each episode depends only on the episode itself

Static vs. dynamic

- Static, the environment is unchanged while an agent is “thinking/considering” about the best action
- The environment is semi-dynamic if the environment itself does not change with the elapse of time, however the agent's performance does change
- The state of dynamic environment and the agent performance change with unknown rule

Discrete vs. Continuous

- A **discrete** environment provides a **limited number** of distinct and clearly defined percepts
- A **continuous** environment generates an infinite number of percepts

Single- vs. multi-agent

- A single agent solves a problem **without interaction** of other agents in its environment
- In a multi-agent system, the agents **interact in cooperation**

Task environment example: Production part-picking robot

- Performance measure, Environment, Actuators, Sensors (PEAS)

Performance	Percentage of parts in correct bins
Environment	Conveyor belt with parts, bins
Actuators	Jointed arm and hand
Sensors	Camera, joint angle sensors

Task environment example : An automated taxi driver

- Performance measure, Environment, Actuators, Sensors (PEAS)

Performance	Safe, fast, legal, comfortable trip, maximize profits
Environment	Roads, other traffic, pedestrians, customers
Actuators	Steering wheel, accelerator, brake, signal, horn
Sensors	Cameras, sonar, speedometer, GPS, odometer, engine sensors, keyboard

Task environment example : Internet shopping agent

- Performance measure, Environment, Actuators, Sensors (PEAS)

Performance	Price, quality, appropriateness, efficiency
Environment	Current and future WWW sites, vendors, shippers
Actuators	Display to user, follow URL, fill in form
Sensors	HTML pages (text, graphics, scripts)

Example: Medical diagnosis system

- Performance measure, Environment, Actuators, Sensors (PEAS)

Performance	Healthy patient, minimize costs, lawsuits
Environment	Patient, hospital, staff
Actuators	Screen display (questions, tests, diagnoses, treatments, referrals)
Sensors	Keyboard (entry of symptoms, findings, patient's answers)

Environment types for Shopping and Taxi examples

- The environment type largely determines the agent design.

	Internet shopping	Taxi
Observable	N	N
Deterministic	Partly	N
Episodic	N	N
Static	Semi	N
Discreate	Y	N
Single-agent	Y (except auctions)	N

Q

Environment types for Shopping and Taxi examples

- The environment type largely determines the agent design.
- What is the environment of the real world?

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Environment types for Shopping and Taxi examples

- The environment type largely determines the agent design.
- What is the

The real world is:

- Partially observable
- Stochastic
- Sequential
- Dynamic
- Continuous
- Multi-agent

Taxi

N

N

N

N

N

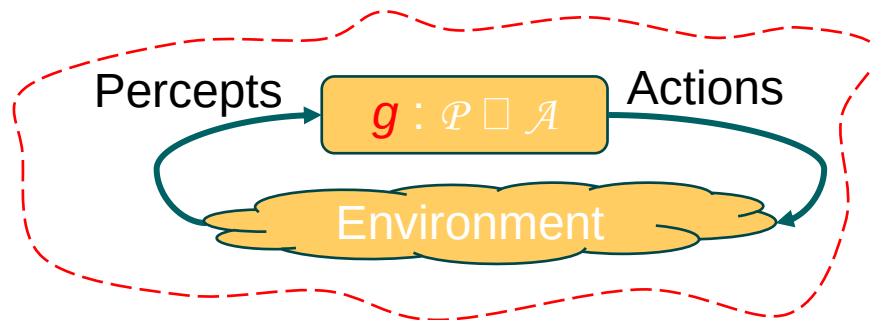
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The structure of agents

Agent Mapping and Structure

- The agent program runs on the computing device – *architecture* – to produce the agent function g
 - The *agent function* $g : \mathcal{P} \rightarrow \mathcal{A}$ maps from percept $\mathcal{P}$ histories to actions $\mathcal{A}$.

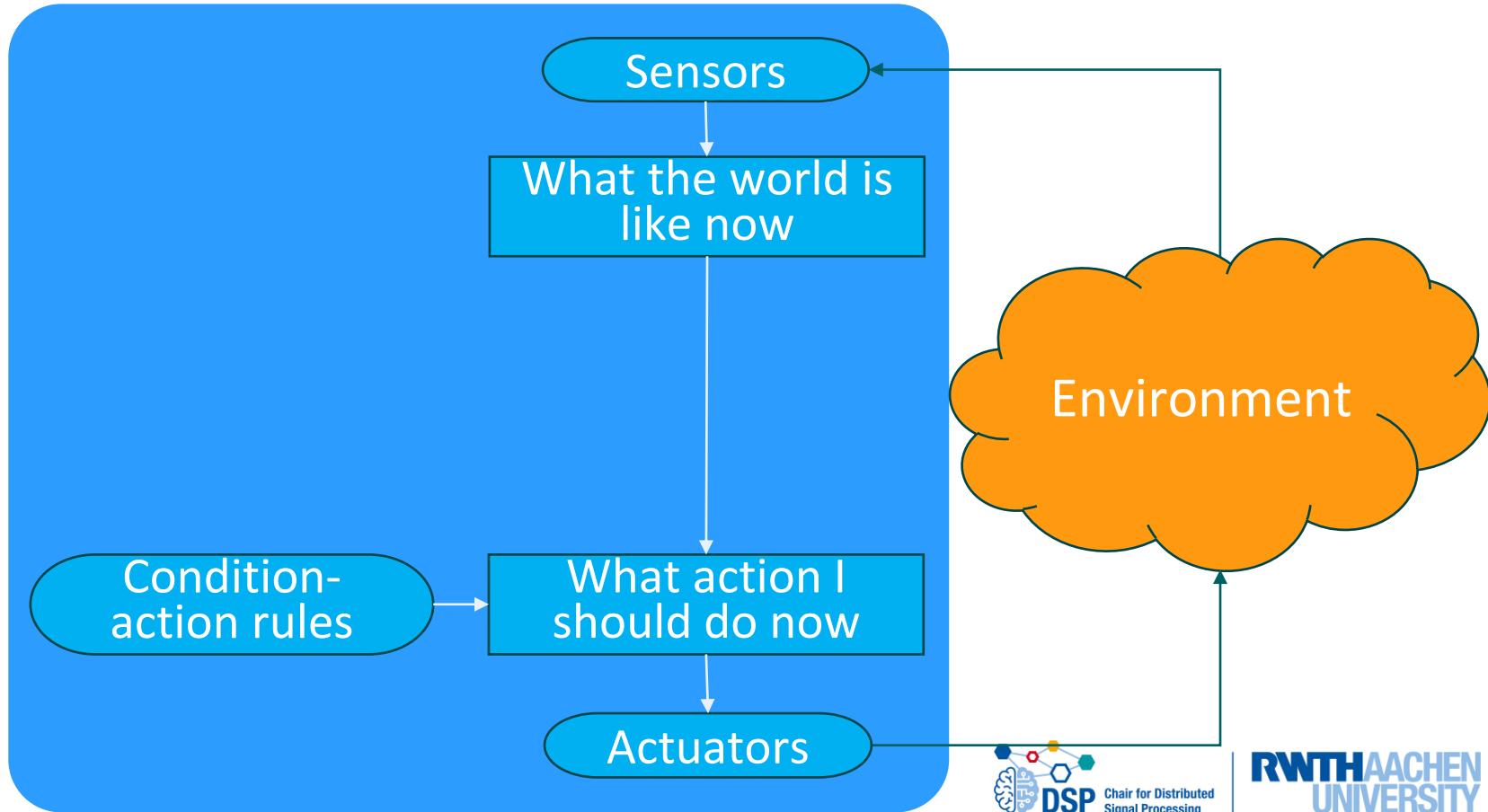
Agent = *Architecture* + Program



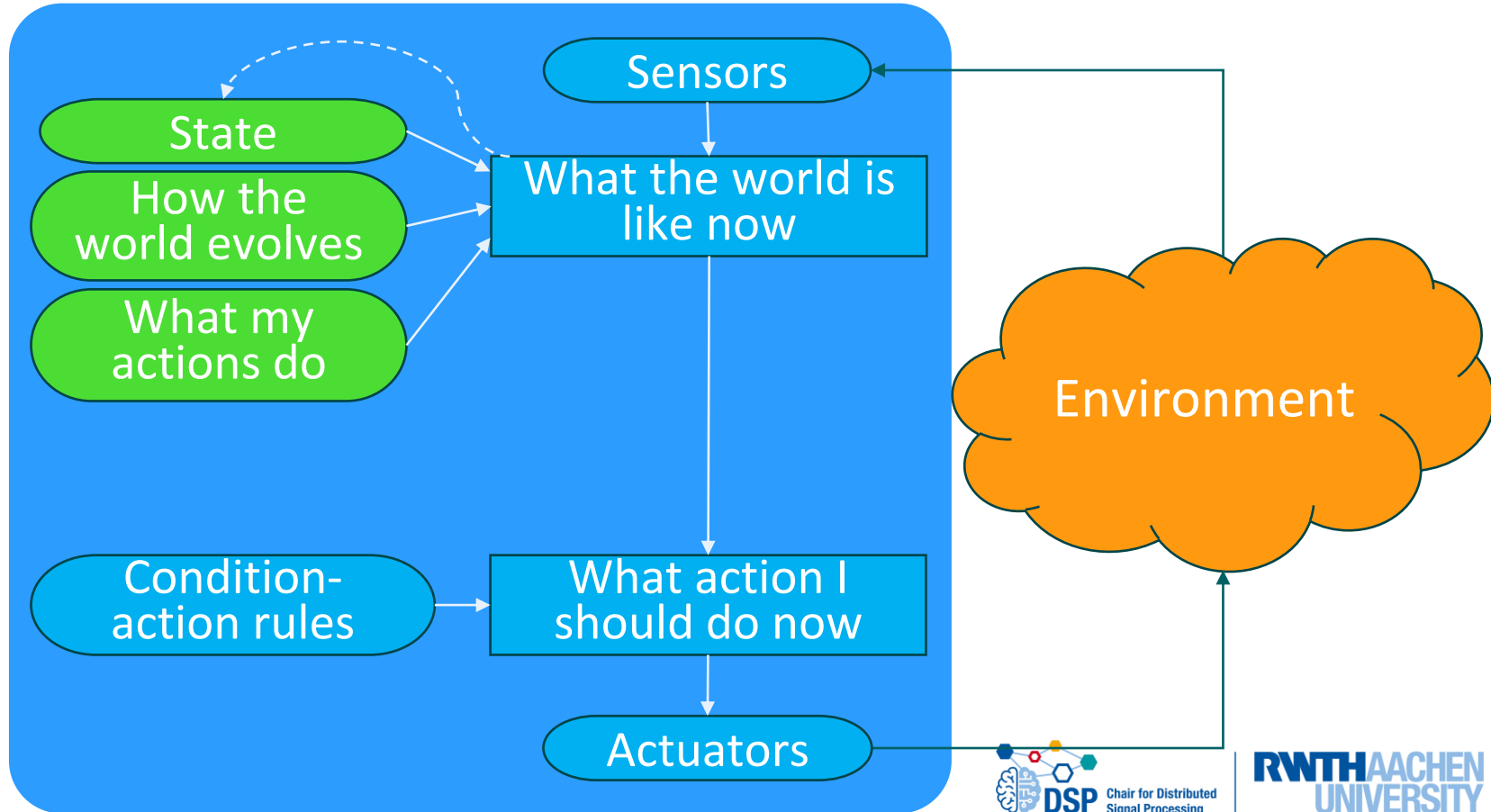
Agent types

- Four basic types of agent architectures are:
 - Simple reflex agents
 - Model-based reflex agents
 - Goal-based agents
 - Utility-based agents
- All these can be turned into learning agents

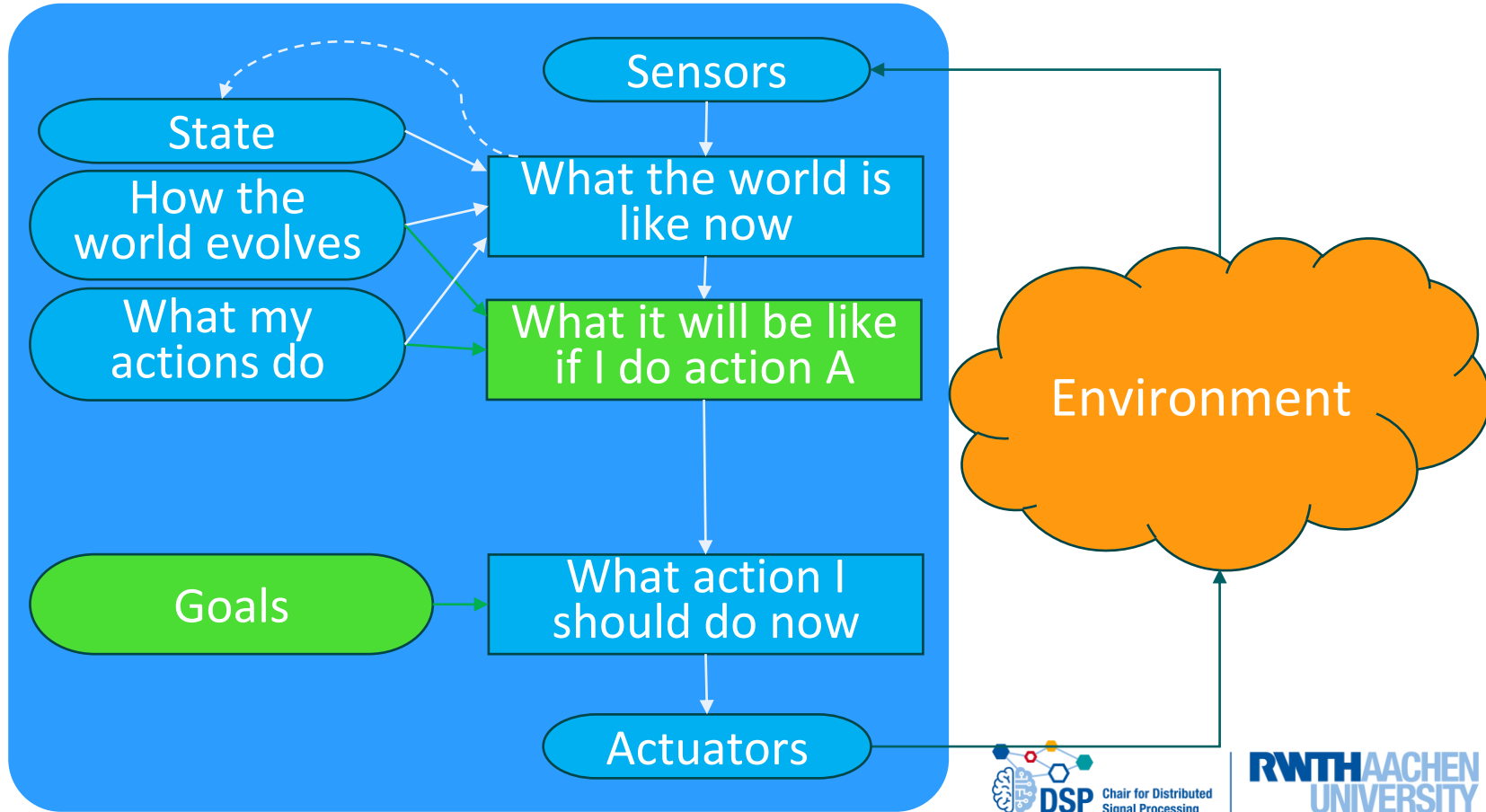
Simplex reflex agents



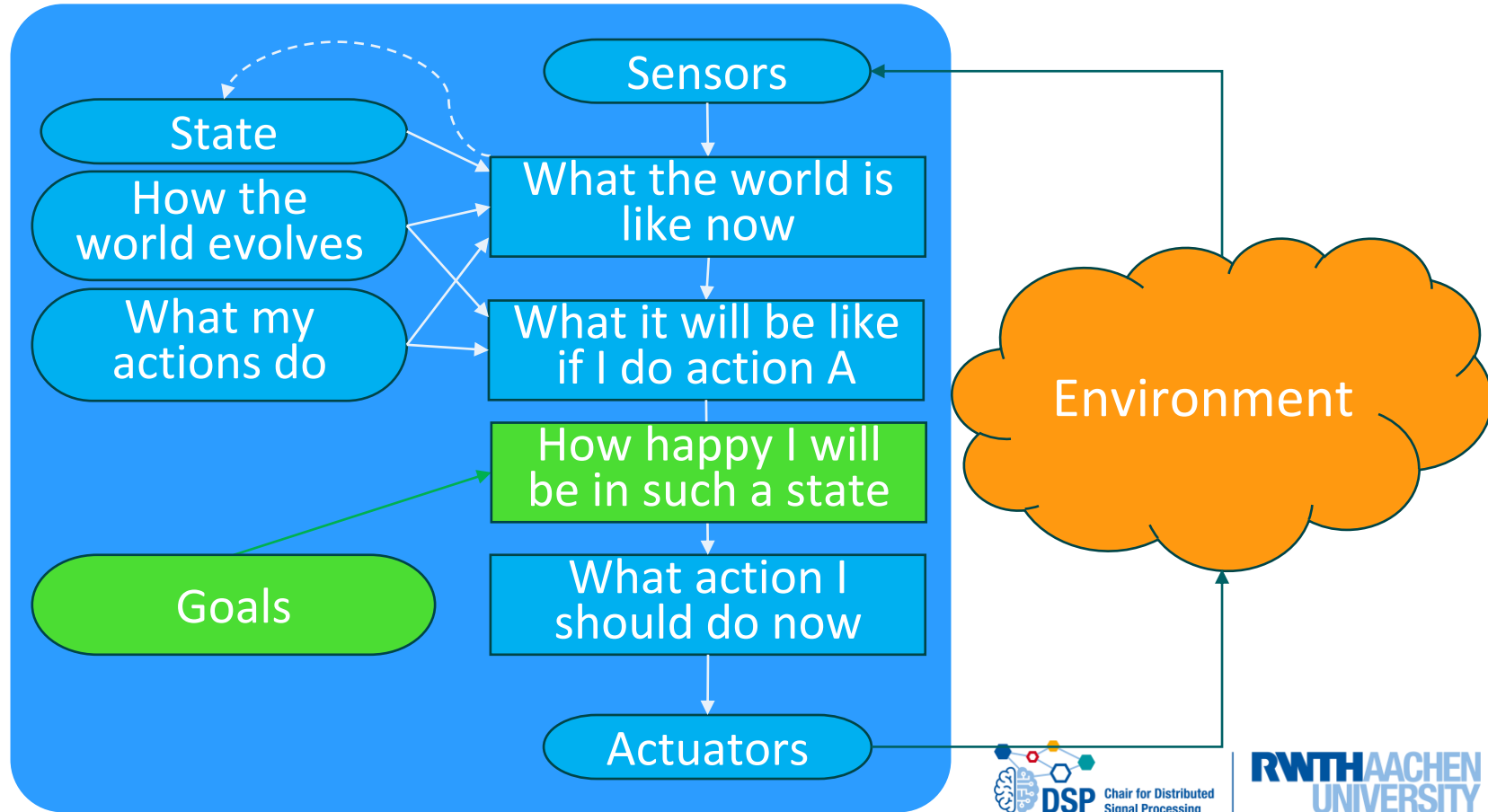
Reflex agents with state



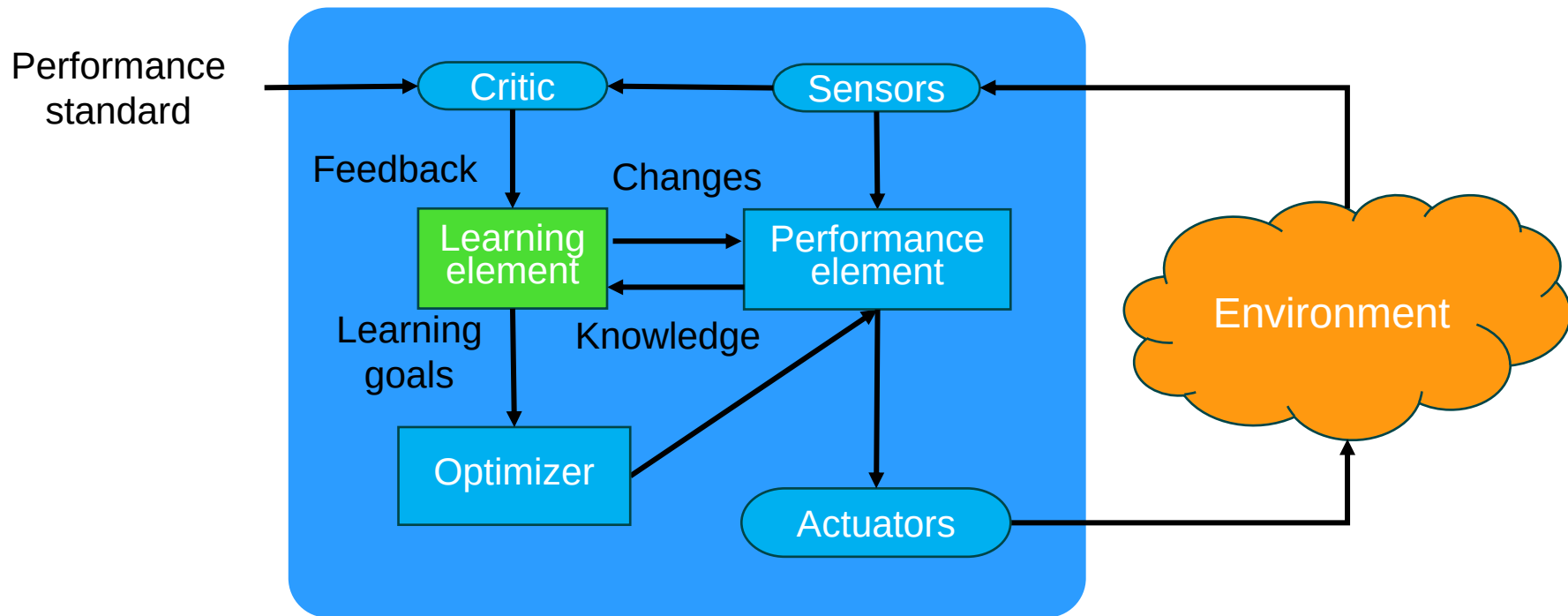
Goal-based agents



Utility-based agents



Learning agents



An agent is learning if it improves its performance on future tasks after making observations about the world.

Learning agent's components to be learned

- A **direct mapping** from the current state to actions

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- A way to infer relevant **properties** of the world from the percept sequence

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- Information on **how the world evolves**
- **Results of possible actions** the agent can take
- **Utility** information indicating the desirability of world states
- **Action-value information** indicating the desirability of actions
- **Goals describing classes of states** whose achievement maximizes the agent's utility

Forms of learning

Feedback to learn from (1/2)

– Supervised learning

- The agent observes some example input (percepts) – output (supervisor) pairs and learns a function that maps enables predictive model testing.
- The output value is available directly from agent's percepts (after the fact); the environment is the teacher.

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– Semi-supervised learning

- The agent is given a few labeled examples and must perform on a large collection of unlabeled examples.

Feedback to learn from (2/2)

– Unsupervised learning

- The agent learns patterns in the input even through no explicit feedback is provided – helps to understand data.
- Clustering is representative: detecting potentially useful clusters of input examples.

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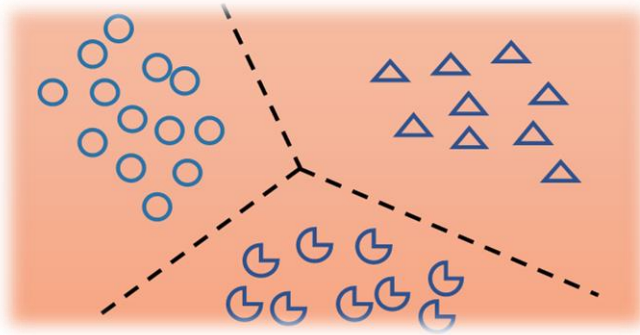
– Reinforcement learning

- The agent learns from a series of reinforcements – rewards or punishments

Learning element

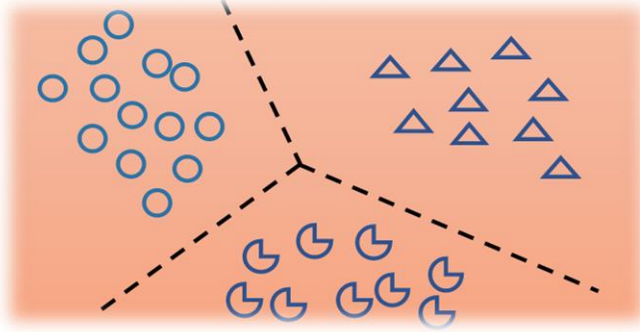
Learning element: Supervised

- Correct answers for each example are provided
- Supervised learning element predicts the value of the function for any valid input after having seen several training examples.



Learning element: Supervised

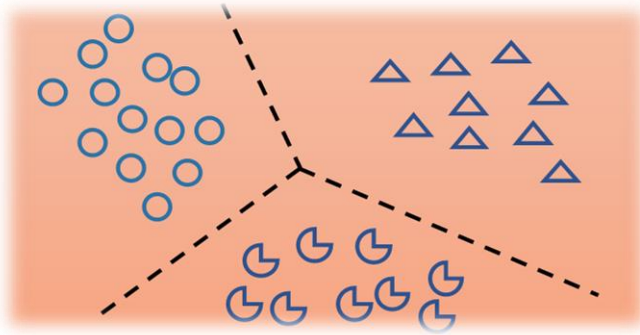
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- If the output is a continuous value, the problem is called **regression**, or **classification** if the output is a discrete value.

Learning element: Supervised

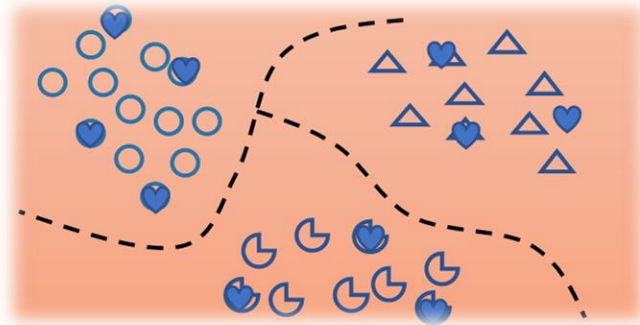
- Correct answers for **each** example are provided



- Supervised learning element predicts the value of the function for any valid input after having seen several training examples.
- If the output is a continuous value, the problem is called **regression**, or **classification** if the output is a discrete value.
- Supervised learning deduces a **function g** from training data (pairs of input data **x** and desired outputs **y**).

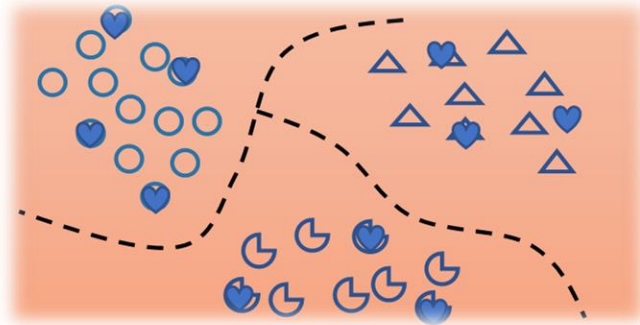
Learning element: Semi-supervised

- Correct answers for **some** examples are provided
- Semi-supervised learning is a technique that makes use of both labeled (a small amount) and unlabeled (a large amount) data for training.



Learning element: Semi-supervised

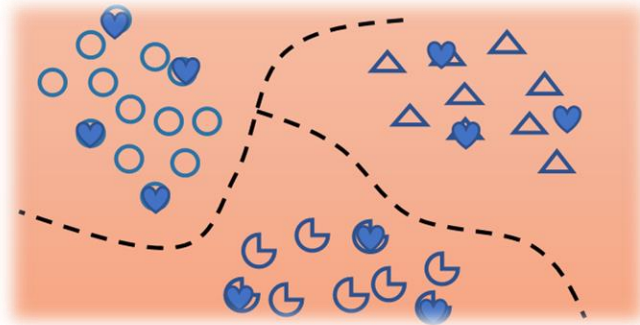
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- Semi-supervised learning is a technique that makes use of both labeled (a small amount) and unlabeled (a large amount) data for training.
- Semi-supervised learning falls between unsupervised learning (without any labeled training data) and supervised learning (with completely labeled training data).

Learning element: Semi-supervised

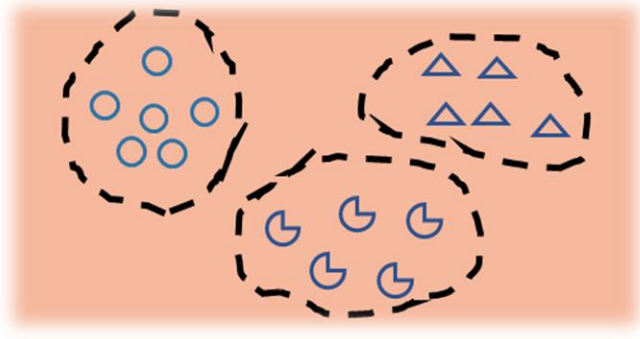
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- Semi-supervised learning is a technique that makes use of both labeled (a small amount) and unlabeled (a large amount) data for training.
- Semi-supervised learning falls between unsupervised learning (without any labeled training data) and supervised learning (with completely labeled training data).
- A small amount of labeled data (domain/expert knowledge) can sometimes produce considerable improvement in learning accuracy.

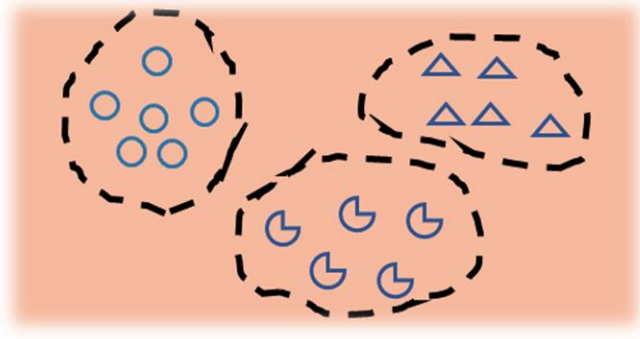
Learning element: Unsupervised

- Correct answers are **not** given
- Unsupervised learning is a class of problems in which one seeks to **determine how data are organized**.



Learning element: Unsupervised

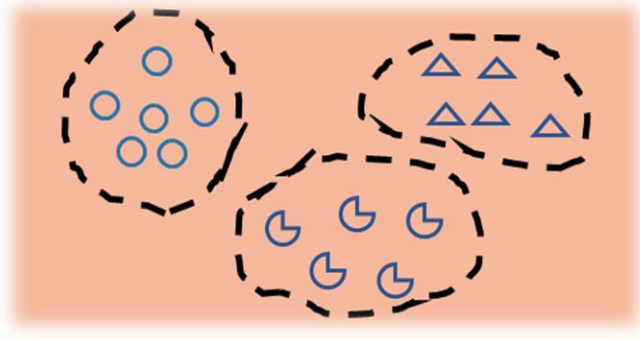
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- Many methods employed here are based on **data mining methods** used to preprocess data.

Learning element: Unsupervised

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- Unsupervised learning is a class of problems in which one seeks to **determine how data are organized**.
- Many methods employed here are based on **data mining methods** used to preprocess data.
- It is distinguished from supervised learning (and reinforcement learning) in that the learner is given only unlabeled examples.

Remarks

- **Agents interact** with environments through actuators and sensors
 - The agent function describes what the agent does in all circumstances
- The **performance measure** evaluates the environment sequence
 - A perfectly rational agent maximizes expected performance
- **Agent programs** implement (some of the) agent functions
- **Environments are categorized along several dimensions**: observable?
deterministic? episodic? static? discrete? single-agent?
- Several basic **agent architectures** are:
 - reflex, reflex with state, goal-based, utility-based, learning

Thank you!



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